

B3093 - MATHEMATICAL AND MACHINE LEARNING METHODS IN IMAGING

Second cycle degree (LM) in Mathematics – A.Y. 2023-2024

Lab 1

1. **2D Image processing: restoration and inpainting.** Complete the script code `MAIN_1_EX1.m` by implementing the following restoration and inpainting tasks:

- **Restoration:** restore an image x corrupted by blur (blur matrix A) and additive white Gaussian noise n according to the degradation model:

$$b = Ax, \quad b = \bar{b} + n,$$

where n is a realization of AWGN. Extend the code with the following features:

- **TIK-L2 method** Compute the restored image x^* by solving the TIK-L2 optimization problem. (without forming matrix A , solve the resulting linear system by discrete FFT)
- **TV-L2 method** Compute the restored image x^* by solving the TV-L2 optimization problem by implementing and applying the gradient descent method `TV_L2_GD.m`.

Compare the restored images obtained by the two methods for different values of the noise standard deviation σ and the parameters characterizing the Gaussian blur (`band` and `sigma`). Determine the optimal regularization parameter for each. Validate the results both by visual inspection and quantitatively by evaluating the relative error (`compute_rel_err.m`) and SNR (using Matlab's `snr()` routine). Plot the relative error as a function of the λ regularization parameter.

- **Inpainting:** inpainting refers to the art of restoring lost parts of image and reconstructing them based on the background information. Compute the reconstructed image x_λ^* by solving the TIK-L2 optimization problem

$$x_\lambda^* = \arg \min_x \|Sx - b\|_2^2 + \lambda \|x\|_2^2,$$

where S is the inpainting mask (a binary diagonal matrix with elements $s_{ii} = 0$ if the pixel i belongs to the inpainting set, 1 otherwise. Validate the results by visual inspection. Create your own inpainting example, providing mask, input image and inpainted result.

2. **1D Signal processing: denoising, deblur and inpainting.** Almost all signal acquisition methods have the effect of perturbing the original signal, either by blurring (smoothing effect on data) or by additive noise; the acquired signal can be also incomplete, i.e. some samples of the signal are lost or corrupted and thus they become useless. Let $x \in R^N$ be a sampled 1D signal and $b \in R^N$ the corresponding measured 1D signal, and thus degraded. In the script `MAIN_1_EX2.m`, solve the following signal processing problems: denoise, deblur and inpainting, by applying the more appropriate numerical optimization method:

- **Denoise:** reconstruct the degraded 1D signal b , corrupted by AWGN, by the generalized TIK-L2 method; that is by minimizing the following quadratic cost function

$$x_\lambda^* = \arg \min_x \|x - b\|_2^2 + \lambda \|Dx\|_2^2,$$

where D is a matrix representing the finite difference discretization of the second-order derivative. Visualize the noisy-free signal x , the perturbed b together with the reconstructed signal x_λ^* .

- **Restoration (noise+blur):** reconstruct the degraded 1D signal b , corrupted by AWGN and Gaussian blur, by the generalized TIK-L2 method; that is by minimizing the following quadratic cost function

$$x_{\lambda}^* = \arg \min_x \|Ax - b\|_2^2 + \lambda \|Dx\|_2^2,$$

where A is the blur matrix, D is a matrix representing the finite difference discretization of the second-order derivative. Visualize the noisy-free signal x , the perturbed b together with the reconstructed signal x_{λ}^* . Evaluate the reconstructions obtained by calculating the relative error of the reconstructed signal with respect to the non-corrupt signal (using `compute_rel_err.m`). Select the regularization parameter according to the discrepancy principle.

- **Inpainting:** consider a signal in which some samples (with random distribution in the signal) are lost or corrupted to the point of being unusable. We want to reconstruct the missing samples on the basis of the available data. Let x be a signal of length N , and suppose to know only K samples of it, where $K < N$. Let b be the vector consisting of known samples of x , obtained by subsampling x :

$$b = Sx$$

where S is the subsampling matrix of dimension $K \times N$. Given S and b determine x defined as

$$x = S^T b + S_c^T v, \tag{1}$$

where S_c contains the rows of the identity matrix that are not in S . Assuming the original signal x is "smooth", determine v in (1) by solving the optimization problem:

$$v^* = \arg \min_v \|D(S^T b + S_c^T v)\|_2^2,$$

where D is a matrix representing the finite difference discretization of the second-order derivative. Finally, reconstruct and display the estimated signal x from (1). Compute the relative error committed. Remark: select `s_id = 10` in the 1D signal generator `GENERATE_DATA_1D.m`.